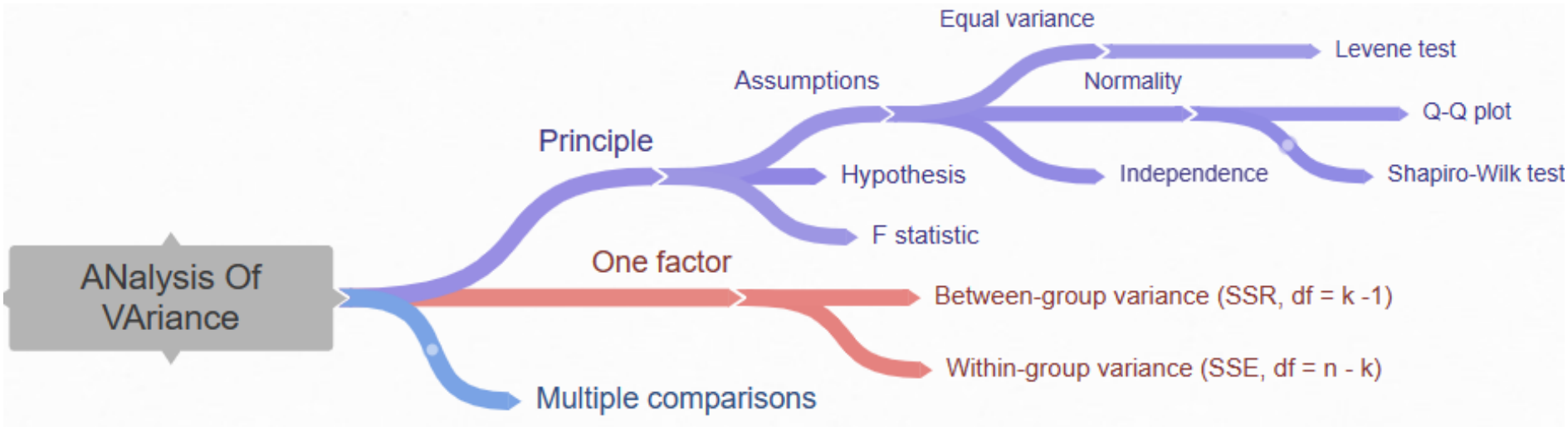
Generalized linear models

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Review1


ANalysis Of VAriance
Principle
Assumptions
Equal variance
Levene test
Normality
Q-Q plot
Shapiro-Wilk test
Independence
Hypothesis
F statistic
One factor
Between-group variance (SSR, df = k -1)
Within-group variance (SSE, df = n - k)
Multiple comparisons

Model comparison

Akaike information criterion (AIC)

$$AIC = 2k - 2log\ell$$

- $2k$: model complexity
- ℓ: likelihood, how likely the model is, given the data

Maximum Likelihood Estimation

Likelihood

- Likelihood function: represents how likely it is to obtain a certain set of observations z_i from a given model (parameter θ)

$$L = f(z_1|\theta)f(z_2|\theta)...f(z_n|\theta) = \prod_{i=1}^{n} f(z_i|\theta)$$

$$log(L) = \sum_{i=1}^{n} log(f(z_i|\theta))$$

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Maximum Likelihood Estimation (MLE)

$$\theta^* = argmax(log(L))$$

to select the parameters θ that make the observed data the most likely

An example

Suppose that we observe a pair of data c(0, 3):

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1. We propose that they come from a Normal distribution
2. Either from $\mathcal{N}(1, 2)$ or $\mathcal{N}(2, 1)$
3. The Log-likelihoods are, respectively:

$$
\begin{aligned}
log(L_1) &= log(f(x_1 = 0, x_2 = 3|\mu = 1, \sigma = 2)) \\
&= log(f(x_1 = 0|\mu = 1, \sigma = 2)) + log(x_2 = 3|\mu = 1, \sigma = 2) \\
&= -3.849 \\
log(L_2) &= log(f(x_1 = 0, x_2 = 3|\mu = 2, \sigma = 1)) \\
&= log(f(x_1 = 0|\mu = 2, \sigma = 1)) + log(x_2 = 3|\mu = 2, \sigma = 1) \\
&= -4.338
\end{aligned}
$$

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\end{aligned}
$$

4. Compared with $\mathcal{N}(2, 1)$, the data is more likely to be generated from $\mathcal{N}(1, 2)$

Another example

假设我们有一枚硬币，我们想知道这枚硬币正面朝上的概率（参数 p）。我们进行了 10 次抛硬币实验，结果正面出现了 6 次，反面出现了 4 次。

1. 定义似然函数
$$L = p^6 * (1-p)^4$$
2. 对数似然
$$logL = 6log(p) + 4log(1-p)$$
3. 最大化对数似然
对 log L 求导，并令导数为零，求解得到 p 的最大似然估计值：
$$p = 0.6$$

Model comparison

Akaike information criterion (AIC)

$$AIC = 2k - 2logL$$

Bayesian Information Criterion (BIC)

$$BIC = log(n)k - 2logL$$

BIC places a higher penalty on the number of parameters in the model (k) so will tend to reward more parsimonious (smaller) models.

When we look back

Y and X

- Categorical Y and Categorical X:

When we look back

Y and X

- Categorical Y and Categorical X:

$$\chi^2$$

- Numerical Y and Categorical X:

When we look back

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ANOVA

- Numerical Y and Numerical X:

When we look back

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ANOVA

- Numerical Y and Numerical X:

Simple Linear, Multiple Linear, polynomial

When we look back

Y and X

- Categorical Y and Categorical X:

$$\chi^2$$

- Numerical Y and Categorical X:

ANOVA

- Numerical Y and Numerical X:

Simple Linear, Multiple Linear, polynomial

- Weird Y: non-normal, even not continuous

?

Genearlized linear model

When the outcome variable is non-normal, even not continuous

1. The outcome variable may be categorical.
   - Binary variables: Passed/failed
   - Polytomous variables: Poor/good/excellent
2. The outcome variable may be a count:

   the number of drinks per day

Methods

1. Logistic model
2. Poisson model

Classification problem

A binary classification case

Questions	Answer y	
Will the student pass the exam?	No	Yes
Is the transaction fraudulent?	No	Yes

- y can only be one of two values
- The classes (or categories) are commonly referred to as No/Yes, or Negative/Positive, or False/True, or simply 0/1
- negative $\neq$ bad, positive $\neq$ good
- negative $\rightarrow$ absence, positive $\rightarrow$ presence

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- p(y=Yes|x)? The most probable category is then chosen as the class for the observation

Logistic model

A binary classification case

To predict Y (Y=0 or Y=1) based on X

- To estimate the probability of Y=1 given X: pr(Y=1|X)
- A decision boundary: e.g., if

$$pr(Y = 1|X) > 0.5$$

then we predict that $\hat{Y} = 1$, otherwise, if:

$$pr(Y = 1|X) < 0.5$$

then we predict that $\hat{Y} = 0$

The need for a link function

Input: X

- Linear model

$$f(x) = \beta_0 + \beta_1 x_1 + .. + \beta_k x_k$$

Output: Y (0 or 1)

- P: $Pr(y = 1)$

$$0 < Pr(y = 1) < 1$$

The need for a link function

Input: X

- Linear model

$$f(x) = \beta_0 + \beta_1 x_1 + .. + \beta_k x_k$$

Output: Y (0 or 1)

- P: $Pr(y = 1)$

$$0 < Pr(y = 1) < 1$$

The link function: (-inf, +inf) $\rightarrow (0, 1)$

$$s(z) = \frac{1}{1 + e^{-z}}$$

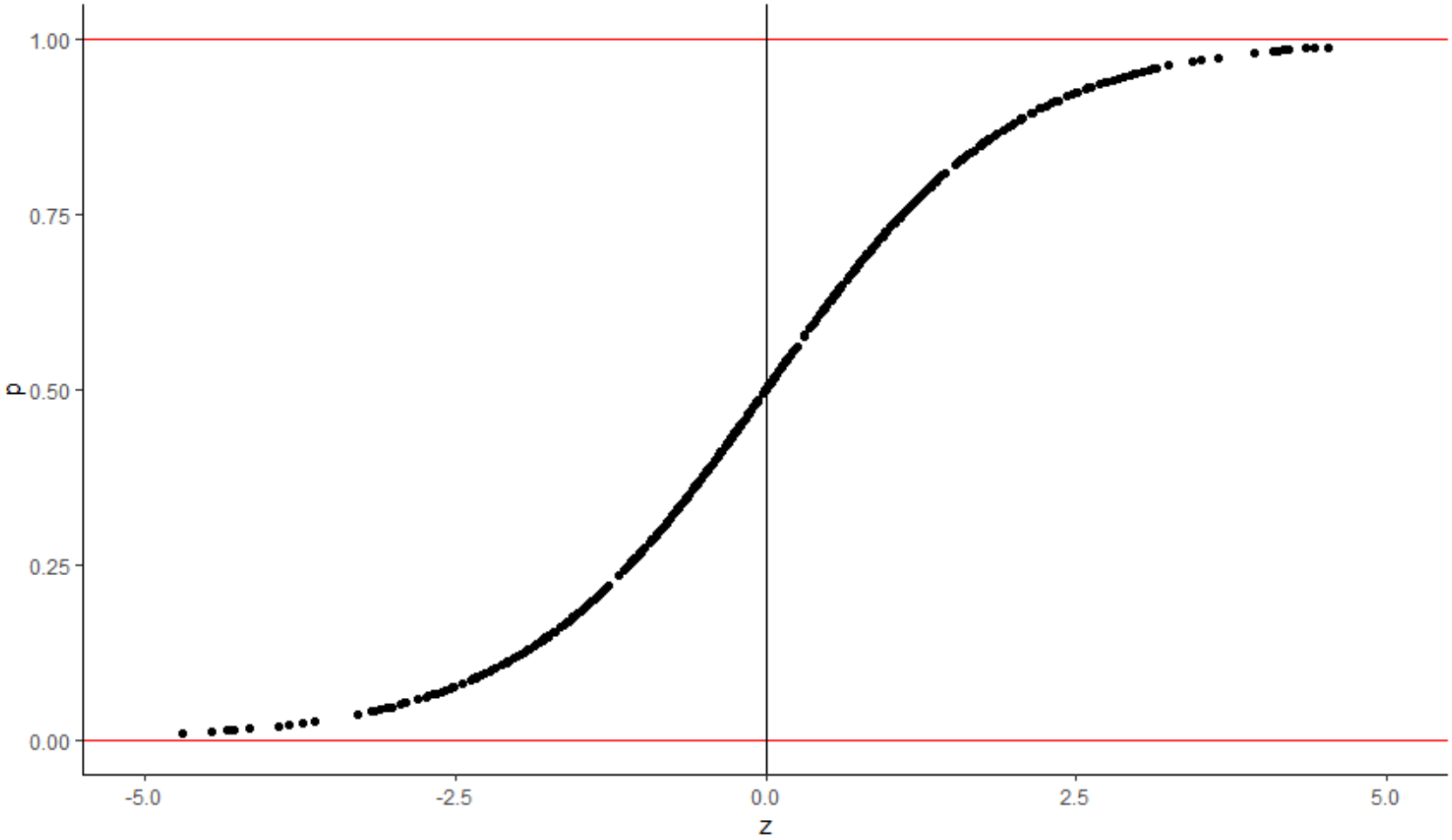
Link function

Sigmoid function

$$s(z) = \frac{1}{1 + e^{-z}}$$

- $z \to -\infty, s(z) \to 0$
- $z \to \infty, s(z) \to 1$
- $z = 0, s(z) = \frac{1}{2}$

Sigmoid function



Logistic model

Input: X

- Linear model

$$f(x) = \beta_0 + \beta_1 x_1 + .. + \beta_k x_k$$

Output: Y (0 or 1)

- Probability of a category

$$pr(y = 1) \in [0, 1]$$

Sigmoid: $s(z) = \frac{1}{1+e^{-z}}$

$$\frac{1}{1 + e^{-f(x)}} = pr(y = 1)$$

- pr(y=1) $>$ 0.5: x is classified to class y $= 1$
- pr(y=1) $<$ 0.5: x is classified to class y $= 0$

Logistic model

From numerical to categorical

1. Numerical x
2. f(x)
3. Probability

$$p = \frac{1}{1 + e^{-f(x)}}$$

4. Category Y

Logistic model

$$p = \frac{1}{1 + e^{-f(x)}}$$

Odds

We can rewrite the above equation as:

$$\frac{p}{1-p} = e^{f(x)}$$

- The quantity $\frac{p}{1-p}$ is called odds, taking any value between 0 and ∞
- E.g., when $p = 0.8$, $odds = 4$

Logistic model

log-odds

$$ln(\frac{p}{1-p}) = f(x)$$
$$= \beta_0 + \beta_1 x$$

- The left-hand side is called the log-odds or logit
- The logit is linear in x

- β_1 gives the change in log odds when increasing X by one-unit
- e^{β_1} gives the change in odds when increasing X by one-unit

Logistic model

Maximum likelihood estimation

- We seek estimates for β_0 and β_1 such that the predicted probability $\hat{p}(y_i = 1)$ (of passing the exam) for each individual, corresponds as closely as possible to the individual's observed status.

MLE for logistic

For an observation (x_i, y_i):

$$L(\beta_0, \beta_1; y_i, x_i) = p^{y_i}(1-p)^{1-y_i}$$

For all observations:

$$L(\beta_0, \beta_1; y, X) = \prod_{i=1}^{n} p^{y_i}(1-p)^{1-y_i}$$

Logistic model

Making predictions

$$ln(\frac{\hat{p}}{1-\hat{p}}) = \hat{\beta}_0 + \hat{\beta}_1 x$$

Decision boundary:

- If $\hat{p} > 0.5$, y is assigned to category 1
- If $\hat{p} < 0.5$, y is assigned to category 0

Note that the threshold is not necessarily always 0.5. We can increase or decrease the bar depending on the situation.

Logistic model

Performance of Logistic model

1. AIC (Akaike Information Criteria)
   - In logistic regression, AIC is the analogous metric of adjusted R^2. We prefer the model with the smallest AIC value.
2. Confusion Matrix
   - A tabular representation of Actual vs Predicted values.

Performance of Logistic model

Confusion matrix

		Predictions (Expectations)		
		0	1	Total
Observations (True values)	0	true negative (TN)	false positive (FP)	N
	1	false negative (FN)	true positive (TP)	P
	Total	N*	P*	

- Accuracy: $\frac{TN+TP}{N+P}$
- True positive rate: $\frac{TP}{P}$
- False positive rate: $\frac{FP}{N}$

Logistic model

Hypothesis tests for a coefficient

	Estimate	Std. error	Z value	Pr(>|z|)	
Intercept	-16.93066	3.23684	-5.231	1.69e-07	***
length	0.43673	0.07993	5.464	4.65e-08	***

- The basic setup is exactly the same as what we have seen before, except we use a Z test.
- However, the Std.error is way beyond the scope of this course.

Logistic model

Testing for the slope of "length"

	Estimate	Std. error	Z value	Pr(>|z|)
Intercept	-16.93066	3.23684	-5.231	1.69e-07 ***
length	0.43673	0.07993	5.464	4.65e-08 ***

$$H_0 : \beta_{length} = 0$$
$$H_1 : \beta_{length} \neq 0$$

$$z = \frac{\hat{\beta}_{length} - \beta_{length}}{SE_{length}} = \frac{0.43673 - 0}{0.07993} = 5.464$$

$$p = P(z > 5.464) + P(z < -5.464) = 4.65e-08$$

Logistic model

Confidence interval for slope coefficient

	Estimate	Std. error	Z value	Pr(>|z|)
Intercept	-16.93066	3.23684	-5.231	1.69e-07 ***
length	0.43673	0.07993	5.464	4.65e-08 ***

Remember, the interpretation for a slope is the change in log odds ratio per unit change in the predictor.

Log odds ratio:

$$CI = 0.43673 \pm 1.96 \times 0.07993 = (0.2801, 0.5934)$$

Odds ratio:

$$e^{CI} = (e^{0.2801}, e^{0.5934}) = (1.3232, 1.8101)$$

When Y has more than two categories

I: Multiple logistic

Y : a, b, or c

1. $$ln(\frac{p_a}{1-p_a}) = f_a(X)$$
$$ln(\frac{p_b}{1-p_b}) = f_b(X)$$
$$ln(\frac{p_c}{1-p_c}) = f_c(X)$$

2. Softmax function

$$p_a^* = \frac{\frac{p_a}{1-p_a}}{\frac{p_a}{1-p_a} + \frac{p_b}{1-p_b} + \frac{p_c}{1-p_c}}; p_b^* = \frac{\frac{p_b}{1-p_b}}{\frac{p_a}{1-p_a} + \frac{p_b}{1-p_b} + \frac{p_c}{1-p_c}}; p_c^* = \frac{\frac{p_c}{1-p_c}}{\frac{p_a}{1-p_a} + \frac{p_b}{1-p_b} + \frac{p_c}{1-p_c}}$$

$$p_a^* + p_b^* + p_c^* = 1$$

When Y has more than two categories

II: Multinomial logistic

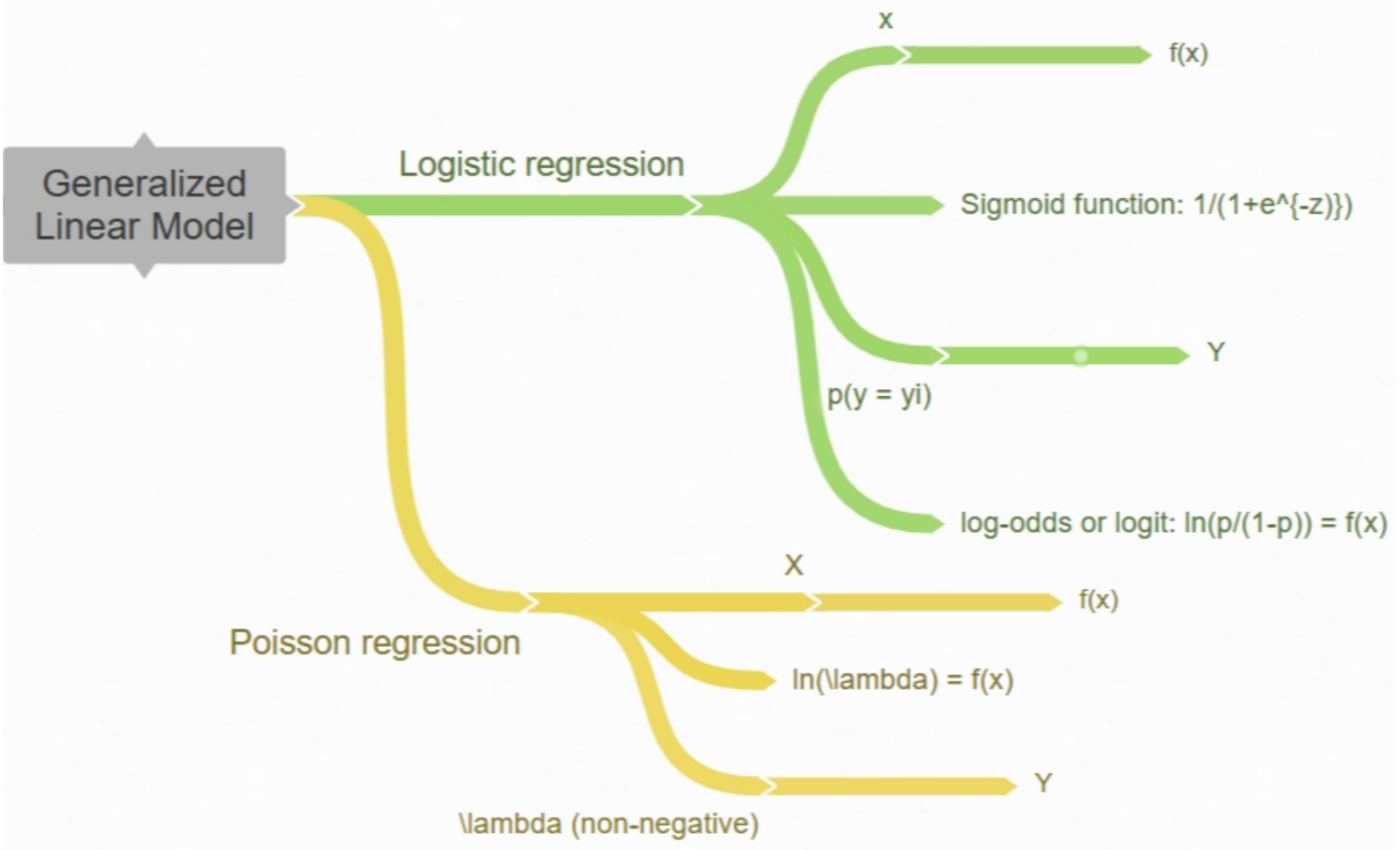
Y : a, b, or c

1. Set a reference category, e.g., when $y = a$
2. Estimate $k - 1$ models (assume that Y has k categories):

$$ln\frac{p_b}{p_a} = \beta_{0b} + \beta_{1b}X$$
$$ln\frac{p_c}{p_a} = \beta_{0c} + \beta_{1c}X$$

3. Relative risk
   The ratio of the probability of choosing one outcome category over the probability of choosing the baseline category is often referred as relative risk

Summary


Generalized Linear Model
Logistic regression
x
f(x)
Sigmoid function: 1/(1+e^{-z})}
Y
p(y = yi)
log-odds or logit: ln(p/(1-p)) = f(x)
Poisson regression
X
f(x)
ln(\lambda) = f(x)
Y
\lambda (non-negative)

Exercise

建立一个逻辑回归模型，预测汽车的传动类型（自动还是手动），并根据模型输出预测的概率。

1. 使用 R 自带的数据集 mtcars，在这个数据集中，am 列表示汽车的传动类型（0 = 自动，1 = 手动）
2. 选择 mpg 和 hp 作为自变量，建立逻辑回归模型预测 am
3. 解释逻辑回归模型的系数和预测概率
4. 使用模型对数据集进行预测，并将预测结果与实际结果进行比较